

Notes: Gowrisankaran, Nevo, and Town (AER, 2015)

Mergers When Prices Are Negotiated: Evidence from the Hospital Industry

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1 Big picture

- **Question.** How should antitrust economists evaluate mergers when prices are set by bilateral bargaining rather than by posted prices or Bertrand competition?
- **Empirical setting.** The paper studies hospital-managed care organization (MCO) negotiations in Northern Virginia. The focal case is the proposed acquisition of Prince William Hospital (PWH) by the Inova hospital system, a transaction challenged by the Federal Trade Commission.
- **Central institutional fact.** Inpatient hospital prices are negotiated between MCOs and hospital systems. Patients pay only a small out-of-pocket share through coinsurance, while MCOs pay the remainder. Therefore patient demand elasticities are much smaller than the elasticities relevant for MCO bargaining.
- **Core model.** The paper builds a two-stage model. In stage 1, MCOs and hospital systems bargain over hospital inclusion and negotiated base prices. In stage 2, patients draw illnesses and choose hospitals from their MCO network, paying coinsurance on negotiated prices.
- **Key economic mechanism.** MCOs bargain partly to reduce insurer costs and partly to preserve enrollee welfare. Because patients are insured, MCO incentives are more price elastic than patient demand; this gives MCOs bargaining leverage against hospitals.
- **Main estimates.** The model implies high hospital price-cost margins: quantity-weighted Lerner indices range from roughly 0.22 to 0.58 across reported hospital systems.
- **Merger result.** The proposed Inova/PWH merger is predicted to raise prices by about 3.1 percent for the combined system, with an increase of about 30.5 percent relative to PWH base revenues.
- **Policy result.** A conduct remedy requiring separate bargaining between Inova and PWH does not meaningfully eliminate the merger price increase.
- **Coinsurance result.** Raising coinsurance tenfold, to levels closer to optimal insurance models with moral hazard, reduces prices by roughly 16 percent.
- **Robustness result.** A richer model in which MCOs compete for enrollees via posted premiums produces similar qualitative conclusions and, for the merger, even larger price effects.
- **Takeaway.** In negotiated-price industries, merger simulation requires modeling the bargaining game, not just demand substitution. In health care, MCO bargaining and patient coinsurance jointly determine how much hospital market power translates into prices.

2 Incremental contribution of this paper

2.1 Relative to standard merger simulation

Traditional merger simulation in industrial organization usually assumes that firms set product prices directly, often under differentiated-products Bertrand competition. That approach is hard to apply to hospitals because patients often pay very little of the negotiated hospital price. If one treated patient demand as the relevant pricing elasticity, the implied Bertrand markups could be implausibly high or even imply negative marginal costs.

Incremental contribution: This paper shows how to do merger simulation when prices are negotiated between upstream providers and downstream insurers. It replaces direct hospital price-setting with bilateral bargaining between MCOs and hospital systems, while retaining the differentiated-demand logic needed to evaluate substitution among hospitals.

2.2 Relative to hospital willingness-to-pay approaches

Earlier work, especially Town and Vistnes and Capps, Dranove, and Satterthwaite, measured hospitals' bargaining value through willingness-to-pay or surplus measures. Those approaches are consistent with bargaining logic, but they do not fully estimate the bargaining game or recover bargaining weights and marginal costs in a structural way.

Incremental contribution: The paper embeds patient willingness-to-pay inside a structural Nash bargaining model. This allows the authors to recover implied marginal costs, bargaining weights, Lerner indices, and counterfactual prices under mergers and conduct remedies.

2.3 Relative to Ho-style hospital–insurer network models

Ho's work emphasizes MCO network formation and downstream health-plan competition. Gowrisankaran, Nevo, and Town study a complementary problem: given observed MCO networks, what prices emerge from bargaining between MCOs and hospital systems?

Incremental contribution: The paper contributes a price-setting model for negotiated hospital prices conditional on networks. It also shows how the negotiated price system can be estimated even without observing the full downstream plan-choice market in the base specification.

2.4 Relative to the bargaining literature

Theoretical bargaining models show that joint negotiation can raise or lower prices depending on the value function and bargaining environment. Empirical applications often require detailed data on both sides of a market.

Incremental contribution: The paper develops an empirically tractable bargaining model in which coinsurance, patient diversion, hospital system ownership, and MCO objectives all affect the Nash bargaining first-order conditions. It derives a markup equation analogous to Bertrand markups but with an additional term capturing MCO bargaining incentives.

2.5 Relative to health-policy debates

Hospital mergers are a major antitrust issue. Policy remedies sometimes require merged hospitals to negotiate separately with payers. Standard arguments often assume that separate bargaining restores competitive leverage.

Incremental contribution: The paper directly evaluates such a conduct remedy in a structural bargaining model and finds that separate bargaining does not solve the problem in this case. The reason is that hospital systems and MCOs still internalize diversion and disagreement effects across related hospitals.

3 Data and measurement

3.1 Market, institutions, and sample

- The market is Northern Virginia, focused on HPD8 and Fauquier County.
- The sample period is 2003–2006, with 2006 used for many counterfactual simulations.
- The main hospitals include Prince William Hospital, Inova Alexandria, Inova Fairfax, Fair Oaks, Loudoun, Mount Vernon, Fauquier, and several non-Inova systems.
- The four MCOs in the data account for about 56 percent of private-pay discharges in this market.
- All 11 in-sample hospitals contract with the four MCOs.

Interpretation: The empirical setting is well suited for merger analysis because there is an actual proposed hospital acquisition, substantial hospital concentration, heterogeneous hospital characteristics, and detailed claims data linking prices, patients, and MCOs.

3.2 Hospital data

- Hospital characteristics come from AHA data and related sources.
- Important variables include mean beds, for-profit status, neonatal intensive care unit availability, and cardiac catheterization lab availability.
- Hospital prices are base prices negotiated between hospitals and MCOs, reported net of disease weights.
- Disease severity is summarized using diagnosis-related-group weights.

Interpretation: The model treats negotiated base prices as prices per unit of disease weight. This makes the hospital price comparable across illnesses and gives the bargaining model a single base-price object for each hospital–MCO pair.

3.3 Patient choice data

- Patient-level hospital choice is estimated using VHI discharge data combined with MCO claims.
- Observed variables include patient demographics, diagnosis weights, hospital choices, travel times, and coinsurance rates.
- Patients can choose an in-network hospital or an outside option.
- Mean coinsurance is low, about 2–3 percent, but it varies across MCOs, hospitals, and illnesses.

Interpretation: Low coinsurance implies that patient hospital choice is not very sensitive to negotiated hospital prices. MCO incentives are therefore essential for disciplining hospital prices.

3.4 Negotiated prices and bargaining variables

- For each MCO–hospital pair, the paper observes negotiated prices from claims data.
- The paper uses observed network participation, prices, and patient choices to estimate demand and the bargaining model.
- Marginal costs are not directly observed; they are recovered from the bargaining first-order conditions.
- Hospital systems may include multiple hospitals, and the bargaining object is between an MCO and a hospital system.

Interpretation: The key empirical advantage is that the authors observe actual negotiated hospital prices and patient allocation, which are precisely the objects needed to estimate bargaining incentives.

4 Model and empirical specifications

4.1 Stage 2: patient hospital choice

Let $j = 1, \dots, J$ index hospitals and $m = 1, \dots, M$ index MCOs. Let $m(i)$ be the MCO of patient i . Patient i with illness d chooses among in-network hospitals $j \in N_{m(i)}$ and an outside option.

The utility from choosing hospital j is

$$u_{ijd} = \beta x_{ijd} - \alpha c_{id} w_d p_{m(i)j} + \varepsilon_{ij}. \quad (1)$$

The outside option is

$$u_{i0d} = \beta x_{i0d} - \alpha c_{id} w_d p_{m(i)0} + \varepsilon_{i0}. \quad (2)$$

- x_{ijd} includes travel time, hospital indicators, and interactions between hospital and patient characteristics.
- c_{id} is the coinsurance rate.
- w_d is the disease weight.
- p_{mj} is the negotiated base price between MCO m and hospital j .
- α is the patient's sensitivity to out-of-pocket price.

Interpretation: Patients react only to their out-of-pocket cost $c_{id} w_d p_{mj}$, not the full negotiated price. This is why patient demand can be inelastic even when MCOs strongly dislike high prices.

4.2 Patient expected welfare

With logit errors, expected utility for patient i with illness d has the usual log-sum form:

$$W_{id}(N_m, p_m) = \log \left[\sum_{j \in N_m} \exp(\beta x_{ijd} - \alpha c_{id} w_d p_{mj}) + \exp(\beta x_{i0d} - \alpha c_{id} w_d p_{m0}) \right]. \quad (3)$$

Aggregating across patients and illnesses gives MCO-level enrollee welfare.

Interpretation: This log-sum object is central to bargaining because hospitals enter MCO networks partly by increasing enrollee surplus.

4.3 MCO objective function

The MCO objective combines enrollee welfare and insurer costs:

$$V_m(N_m, p_m) = \tau W_m(N_m, p_m) - TC_m(N_m, p_m), \quad (4)$$

where τ is the relative weight placed on enrollee welfare.

- If $\tau = 1$, MCOs weight enrollee welfare and insurer costs equally.
- If $\tau < 1$, MCOs put relatively more weight on insurer costs.
- If $\tau > 1$, MCOs put relatively more weight on enrollee welfare.

Interpretation: The estimated τ tells us how strongly MCOs bargain on behalf of enrollee welfare versus cost savings. The estimates imply that MCOs place substantial weight on enrollee welfare.

4.4 Hospital system payoff

The quantity of weighted admissions to hospital j from MCO m is

$$q_{mj}(N_m, p_m) = \sum_i \sum_d 1\{m(i) = m\} f_{id} w_d s_{ijd}(N_m, p_m), \quad (5)$$

where s_{ijd} is the logit choice probability.

Hospital system s earns

$$\pi_s(M_s, \{p_m\}_{m \in M_s}) = \sum_{m \in M_s} \sum_{j \in J_s} q_{mj}(p_{mj} - m c_{mj}). \quad (6)$$

Marginal cost is parameterized as

$$mc_{mj} = \gamma v_{mj} + e_{mj}, \quad (7)$$

where v_{mj} includes hospital, year, and MCO fixed effects.

Interpretation: The hospital payoff is the profit or perceived surplus from serving MCO enrollees. For not-for-profit hospitals, marginal cost can be interpreted as perceived marginal cost net of any utility from providing care.

4.5 Nash bargaining problem

For hospital system s and MCO m , the Nash bargaining product is

$$NB_{m,s}(p_{mj}, p_{m,-j}) = \left[\sum_{j \in J_s} q_{mj}(p_{mj} - mc_{mj}) \right]^{b_{s(m)}} [V_m(N_m, p_m) - V_m(N_m \setminus J_s, p_m)]^{b_{m(s)}}. \quad (8)$$

The bargaining weights satisfy

$$b_{s(m)} + b_{m(s)} = 1. \quad (9)$$

Interpretation: The MCO's disagreement payoff is the value of excluding the hospital system from its network. The hospital system's disagreement payoff is the loss of profits from losing that MCO's patients.

4.6 Equilibrium markup equation

The first-order conditions can be written as

$$q + \Omega(p - mc) = -\Lambda(p - mc), \quad (10)$$

which yields

$$p = mc - (\Omega + \Lambda)^{-1}q. \quad (11)$$

Interpretation: This resembles the markup equation in differentiated-products Bertrand models, but the extra Λ term captures how MCO surplus changes with hospital prices. Bargaining incentives and coinsurance enter through this term.

4.7 Econometric estimation

1. Estimate patient demand using hospital choices, travel times, patient characteristics, disease weights, negotiated prices, and coinsurance.
2. Use observed prices and quantities together with bargaining first-order conditions to recover marginal-cost residuals.
3. Estimate bargaining parameters and cost parameters by GMM, using moments that make cost residuals orthogonal to instruments and fixed effects.
4. Bootstrap standard errors using resampling at the payer/year/system level.

Interpretation: Demand identifies patient substitution. Bargaining identifies hospital costs, bargaining weights, and the way MCO objectives restrain prices.

5 Identification and empirical design

5.1 Demand identification

- Patient hospital choice depends on travel time, hospital characteristics, disease severity, and out-of-pocket price.
- Coinsurance variation helps identify the price coefficient α .
- Hospital-year effects and hospital indicators interacted with disease weight absorb many quality differences.
- The outside option absorbs choices outside the in-sample hospital market.

Interpretation: The demand model primarily uses within-market variation in travel times, hospital attributes, and out-of-pocket prices to identify hospital substitution patterns.

5.2 Bargaining identification

- Observed prices vary across hospitals, MCOs, and time.
- Given demand derivatives, the bargaining first-order conditions imply marginal costs as a function of prices and bargaining parameters.
- Cost fixed effects and instruments restrict which price variation can be attributed to costs versus bargaining power.
- The model is disciplined by the fact that the same hospitals bargain with multiple MCOs at different prices.

Interpretation: Cross-MCO price variation is crucial: it lets the paper separate cost heterogeneity, MCO bargaining weights, and MCO willingness-to-pay differences.

5.3 Incremental credibility of counterfactuals

- The actual Inova/PWH merger proposal provides a real policy counterfactual rather than an abstract merger experiment.
- The model simulates the exact hospital-system ownership changes relevant to the case.
- Conduct remedies can be explicitly represented by changing disagreement values and bargaining structure.
- Coinsurance counterfactuals allow the model to distinguish patient price sensitivity from MCO price sensitivity.

Interpretation: The counterfactuals are not merely reduced-form extrapolations. They impose the estimated bargaining equilibrium under new ownership or insurance designs.

5.4 Limitations

- The base model treats MCO enrollment as fixed and does not model downstream MCO premium competition.
- MCO networks are taken as given.
- Marginal costs are inferred rather than observed directly.
- The market is a high-concentration Northern Virginia case, so magnitudes may differ elsewhere.
- The posted-premium model is calibrated rather than fully estimated.

Interpretation: The paper is strongest for pricing conditional on observed networks and MCO enrollment. It is less definitive about long-run network formation and plan enrollment.

6 Tables and figures

6.1 Figure 1 and Table 1: hospital geography and hospital characteristics

Read:

- Figure 1 maps Northern Virginia hospitals and shows the location of PWH relative to Inova hospitals and other competitors.
- The two closest hospitals to PWH are Inova hospitals, Fair Oaks and Fairfax, which is important for the merger counterfactual.
- Table 1 shows large price variation across hospitals. Mean base prices range from roughly 9,545 to 13,270.
- Hospitals also differ in size, for-profit status, NICU presence, and cardiac catheterization services.

Economic message: Geographic proximity and differentiated hospital characteristics determine substitution patterns. These features shape the bargaining consequences of a merger.

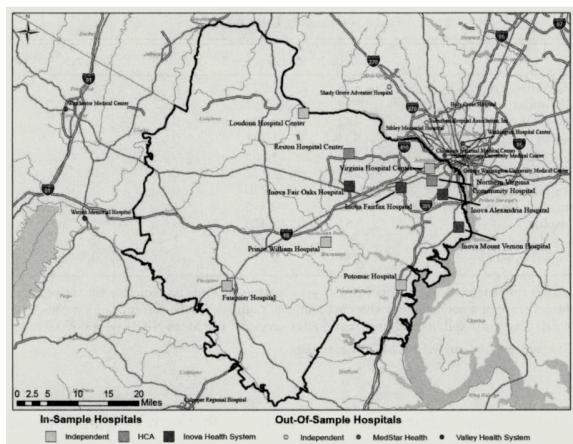


FIGURE 1. 2003 NORTHERN VIRGINIA HOSPITAL LOCATIONS

Figure 1: Northern Virginia hospital locations

TABLE 1—HOSPITAL CHARACTERISTICS

Hospital	Mean beds	Mean price \$	Mean FP	Mean NICU	Mean cath lab
Prince William Hospital	170	10,273	0	1	0
Alexandria Hospital	318	9,757	0	1	1
Fair Oaks Hospital	182	9,799	0	0.5	1
Fairfax Hospital	833	11,881	0	1	1
Loudoun Hospital	155	11,565	0	0	1
Mount Vernon Hospital	237	12,112	0	0	1
Fauquier Hospital	86	13,270	0	0	0
N. VA Community Hosp.	164	9,545	1	0	1
Potomac Hospital	153	11,420	0	1	1
Reston Hospital Center	187	9,973	1	1	1
Virginia Hospital Center	334	9,545	0	0.5	1

Notes: We report (unweighted) mean prices across year and payor. FP is an indicator for for-profit status, Mean NICU for the presence of a neonatal intensive care unit, and Cath lab for the presence of a cardiac catheterization

Table 1: hospital characteristics

6.2 Table 2 and Table 3: patient sample and demand estimates

Read:

- Table 2 shows meaningful heterogeneity in patient age, white share, disease weight, travel time, coinsurance, and hospital shares.
- Inova Fairfax has the largest discharge share among individual hospitals and has high disease weight.
- Table 3 shows that out-of-pocket price, travel time, and travel-time interactions are important for hospital choice.
- The base price coefficient is negative and significant, but the implied own-price elasticities are small because coinsurance rates are low.

Economic message: Demand estimates confirm that patients dislike higher out-of-pocket costs and longer travel, but patient price sensitivity alone is too low to discipline hospital markups.

TABLE 2—PATIENT SAMPLE

Hospital	Mean age	Share white	Mean DRG weight	Mean travel time	Mean coins. rate	Discharges total	Share
Prince William	36.1	0.73	0.82	13.06	0.032	9,681	0.066
Alexandria Hosp.	39.3	0.62	0.92	12.78	0.025	15,622	0.107
Fair Oaks Hosp.	37.7	0.54	0.94	17.75	0.023	17,073	0.117
Fairfax Hospital	35.8	0.58	1.20	18.97	0.023	46,428	0.319
Loudoun Hospital	37.2	0.74	0.81	15.54	0.023	10,441	0.072
Mt. Vernon Hosp.	50.3	0.66	1.38	16.18	0.022	3,749	0.026
Fauquier Hospital	40.5	0.90	0.92	15.29	0.033	3,111	0.021
N. VA Community Hosp.	47.2	0.48	1.43	16.02	0.016	531	0.004
Potomac Hospital	37.5	0.60	0.93	9.62	0.024	8,737	0.060
Reston Hosp. Ctr.	36.8	0.69	0.90	15.35	0.021	16,007	0.110
VA Hosp. Center	40.8	0.59	0.98	15.88	0.017	12,246	0.084
Outside option	39.3	0.82	1.39	0.00	0.029	2,113	0.014
All Inova	37.5	0.59	1.09	17.37	0.024	85,540	0.641
All others	38.1	0.68	0.92	13.74	0.023	60,199	0.359

Note: Mean travel time is measured in minutes.

Sources: Authors' analysis of VHI discharge data and MCO claims data.

Table 2: patient sample

TABLE 3—MULTINOMIAL LOGIT DEMAND ESTIMATES

Variable	Coefficient	Standard error
Base price $\times$ weight $\times$ coinsurance	-0.0008**	(0.0001)
Travel time	-0.1150**	(0.0026)
Travel time squared	-0.0002**	(0.0000)
Closest	0.2845**	(0.0114)
Travel time $\times$ beds/100	-0.0118**	(0.0008)
Travel time $\times$ age/100	-0.0444**	(0.0023)
Travel time $\times$ FP	0.0157**	(0.0011)
Travel time $\times$ teach	0.028**	(0.0010)
Travel time $\times$ residents/beds	0.0006**	(0.0000)
Travel time $\times$ income/1000	0.0002**	(0.0000)
Travel time $\times$ male	-0.0151**	(0.0007)
Travel time $\times$ age 60+	-0.0017	(0.0013)
Travel time $\times$ weight/1000	11.4723**	(0.4125)
Cardiac major diagnostic class $\times$ cath lab	0.2036**	(0.0409)
Obstetric major diagnostic class $\times$ NICU	0.6187**	(0.0170)
Nerv. circ. musc major diagnostic classes $\times$ MRI	-0.1409**	(0.0460)

Notes: Specification also includes hospital-year interactions and hospital dummies interacted with disease weight. Pseudo $R^2 = 0.445$, $N = 1,710,801$.

**Significant at the 1 percent level.

Table 3: multinomial logit demand estimates

6.3 Table 4 and Table 5: demand elasticities and bargaining parameters

Read:

- Table 4 reports own-price elasticities between about -0.098 and -0.153 for the selected hospitals.
- These elasticities are much less than one in absolute value, implying very inelastic patient demand.
- Table 5 estimates the bargaining model. In specification 1, the MCO welfare weight τ is 2.79, and hospital and MCO bargaining weights are fixed at 0.5.
- In specification 2, MCO 2 and 3 have essentially all bargaining weight, while MCO 4 has a lower but still high bargaining weight.
- Estimated hospital cost parameters vary substantially across hospitals.

Economic message: Patient demand alone cannot explain observed prices. The bargaining model adds MCO incentives and cost heterogeneity, allowing the authors to recover plausible margins and counterfactuals.

TABLE 4—MEAN ESTIMATED 2006 DEMAND ELASTICITIES FOR SELECTED HOSPITALS

Hospital	PW (1)	Fairfax (2)	Reston (3)	Loudoun (4)	Fauquier (5)
1. Prince William	-0.125	0.052	0.012	0.004	0.012
2. Inova Fairfax	0.011	-0.141	0.018	0.006	0.004
3. HCA Reston	0.008	0.055	-0.149	0.022	0.002
4. Inova Loudoun	0.004	0.032	0.037	-0.098	0.001
5. Fauquier	0.026	0.041	0.006	0.002	-0.153
6. Outside option	0.025	0.090	0.022	0.023	0.050

Note: Elasticity is $\frac{\partial \ln q_{ij}}{\partial p_j}$ where j denotes row and k denotes column.

Table 4: estimated demand elasticities

TABLE 5—BARGAINING PARAMETER ESTIMATES

Parameter	Specification 1		Specification 2	
	Estimate	SE	Estimate	SE
MCO welfare weight (τ)	2.79	(2.87)	6.69	(5.53)
MCO 1 bargaining weight	0.5	—	0.52	(0.09)
MCOs 2 & 3 bargaining weight	0.5	—	1.00**	(7.77 $\times 10^{-10}$)
MCO 4 bargaining weight	0.5	—	0.76**	(0.09)
<i>Hospital cost parameters</i>				
Prince William Hospital	8,635**	(3,009)	5,971**	(1,236)
Inova Alexandria	10,412*	(4,415)	6,487**	(1,905)
Inova Fairfax	10,786**	(3,765)	6,133**	(1,211)
Inova Fair Oaks	11,192**	(3,239)	6,970**	(2,352)
Inova Loudoun	12,014**	(3,188)	8,167**	(1,145)
Inova Mount Vernon	10,294*	(5,170)	4,658	(3,412)
Fauquier Hospital	14,553**	(3,390)	9,041**	(1,905)
No. VA Community Hosp.	10,086**	(2,413)	5,754**	(2,162)
Potomac Hospital	11,459**	(2,703)	7,653**	(902)
Reston Hospital Center	8,249**	(3,064)	5,756**	(1,607)
Virginia Hospital Center	7,993**	(2,139)	5,303**	(1,226)
Patients from MCO 2	-9,043**	(2,831)	—	—
Patients from MCO 3	-8,910**	(3,128)	—	—
Patients from MCO 4	-4,476	(2,707)	—	—
Year 2004	1.130	(1,303)	1,414	(1,410)
Year 2005	1.808	(1,481)	1,737	(1,264)
Year 2006	1.908	(1,259)	2,459*	(1,077)

Table 5: bargaining parameter estimates

6.4 Table 6 and Table 7: markups and merger counterfactuals

Read:

- Table 6 reports high Lerner indices, from 0.22 at Fauquier Hospital to 0.58 at Virginia Hospital Center.
- Actual patient own-price elasticities are low, but effective own-price elasticities faced by MCOs are much higher.

- Table 7 shows the proposed Inova/PWH merger raises prices by 3.1 percent for the combined system and raises profits by 9.3 percent.
- Relative to PWH base revenue, the price increase is 30.5 percent and the profit increase is 91.5 percent.
- Separate bargaining has very similar price effects, while breaking up Inova lowers prices by 6.8 percent.

Economic message: MCO bargaining substantially restrains prices, but hospital mergers still raise bargaining leverage. Conduct remedies based on separate bargaining do not eliminate the merger problem.

TABLE 6—LERNER INDICES AND ACTUAL AND EFFECTIVE PRICE ELASTICITIES

System name	Lerner index	Actual own price elasticity	Effective own price elasticity	Own price elasticity without insurance
Prince William Hospital	0.52	0.13	1.94	5.16
Inova Health System	0.39	0.07	2.55	3.10
Fauquier Hospital	0.22	0.17	4.56	6.11
HCA (Reston Hospital)	0.35	0.15	2.87	7.34
Potomac Hospital	0.37	0.15	2.74	6.77
Virginia Hospital Center	0.58	0.13	1.74	6.43

Note: Reported elasticities and Lerner indices use quantity weights.

Table 6: Lerner indices and elasticities

TABLE 7—IMPACT OF COUNTERFACTUAL INDUSTRY STRUCTURES

Counterfactual	System	Percent Δ		
		Price	Quantity	Profits
1. Inova/PWH merger	Inova & PWH rival hospitals change at Inova+PW relative to PW base	3.1	-0.5	9.3
		3.6	1.2	12.0
		30.5	-4.9	91.5
2. Inova/PWH merger with separate bargaining	Inova & PWH rival hospitals	3.3	-0.5	8.8
		3.5	1.2	11.2
3. Loudoun demerger	Inova & Loudoun rival hospitals change at Inova relative to Loudoun base	-1.8	0.1	-4.7
		-1.6	-0.2	-4.7
		-14.7	0.8	-38.5
4. Breaking up Inova	All hospitals	-6.8	0.05	-18.9

Table 7: industry-structure counterfactuals

6.5 Table 8 and Table 9: coinsurance and posted premium model baseline

Read:

- Table 8 shows that eliminating coinsurance raises hospital prices by 3.7 percent.
- Raising coinsurance tenfold lowers prices by 16.1 percent and slightly increases quantity.
- The Inova/PWH merger effect is smaller under zero coinsurance than under positive coinsurance, showing that steering incentives interact with market power.
- Table 9 calibrates the posted-premium competition model. Hospital prices are somewhat lower than in the base model, while hospital margins remain high.
- The posted-premium model generates a health insurance take-up rate of 84.5 percent and positive consumer surplus from insurance.

Economic message: Coinsurance matters because it gives MCOs a tool to steer patients. Insurance design and hospital market power interact in determining negotiated prices.

TABLE 8—IMPACT OF COUNTERFACTUAL COINSURANCE LEVELS

Counterfactual	System	% Δ Price	% Δ Quantity	% Δ Profits
1. No coinsurance	All hospitals	3.7	0.01	9.8
2. Coinsurance 10 times current	All hospitals	-16.1	0.9	-0.4
3. Inova/PWH merger, no coinsurance	Inova and PWH	2.9	0	7.4
	rival hospitals	1.3	0	3.9

Note: Price changes are calculated using quantity weights.

Table 8: counterfactual coinsurance levels

TABLE 9—BASE SIMULATION RESULTS FROM POSTED PREMIUM COMPETITION MODEL

Variable	Mean value in posted premium competition model	Mean value in base model
Hospital prices	\$11,088	\$13,618
Hospital margin per patient	\$4,796	\$4,893
MCO premiums	\$1,706	—
MCO margin per enrollee	\$792	—
Consumer surplus	\$4,398	—
Health insurance take-up (%)	84.5	—

Note: Hospital prices are patient-weighted base prices excluding the outside good, hospital margins are patient-weighted, MCO premiums and margins are enrollee-weighted, and consumer surplus is per capita.

Table 9: posted-premium model baseline

6.6 Table 10: merger effects in the posted premium model

Read:

- Table 10 shows that the posted-premium model predicts larger Inova/PWH merger price effects than the base model.
- Inova/PWH prices rise by 7.2 percent, compared with 3.1 percent in the base model.

- MCO premiums rise by 3.4 percent.
- Consumer surplus falls by 4.4 percent and health insurance take-up falls by 1.6 percent.

Economic message: Allowing MCOs to set premiums and compete downstream does not overturn the main findings. If anything, it strengthens the conclusion that the merger harms consumers.

TABLE 10—MEAN INOVA/PWH MERGER EFFECTS FROM POSTED PREMIUM COMPETITION MODEL (*Percent*)

	Posted premium competition model	Base model – specification 1
Inova/PWH prices	7.2	3.1
Other hospitals prices	2.2	3.6
Inova/PWH margin per patient	16.9	9.8
Other hospitals margin per patient	6.6	10.7
MCO premiums	3.4	—
MCO margin per enrollee	1.0	—
Consumer surplus	–4.4	—
Health insurance take-up	–1.6	—

Note: Hospital prices are patient-weighted base prices excluding the outside good, hospital margins are patient-weighted, MCO premiums are enrollee-weighted, MCO margins are enrollee-weighted, and consumer surplus is patient-weighted.

7 Short summary

- The paper estimates a structural bargaining model of negotiated hospital prices.
- It studies Northern Virginia hospitals and MCOs around the proposed Inova/PWH merger.
- Patient choice is estimated with a logit model using travel time, hospital characteristics, disease weights, prices, and coinsurance.
- Patients are very price inelastic because coinsurance rates are low.
- MCOs have stronger incentives than patients to discipline hospital prices because MCOs pay most of the bill.
- The bargaining model implies high hospital margins but also substantial MCO bargaining restraint.
- The proposed Inova/PWH merger raises prices materially.
- Separate bargaining as a conduct remedy does not eliminate the price increase.
- Higher coinsurance lowers hospital prices by strengthening patient and MCO steering incentives.
- A posted-premium competition extension produces qualitatively similar conclusions.
- The key methodological lesson is that merger simulation in negotiated-price industries must model bargaining, not only consumer substitution.

8 Conclusion

8.1 Core conclusion

Gowrisankaran, Nevo, and Town show that negotiated hospital prices are shaped by a three-way interaction among patient demand, MCO bargaining incentives, and hospital system ownership. The proposed Inova/PWH merger would increase prices, and a separate-bargaining conduct remedy would not meaningfully prevent the price increase.

8.2 Economic intuition

Patients pay only a small coinsurance share, so they are not very responsive to hospital prices. MCOs, however, pay most of the cost and therefore bargain aggressively with hospitals. A merger changes the disagreement values in these negotiations: losing access to a merged hospital system becomes more costly to MCOs, increasing hospital leverage. Coinsurance changes the ability of MCOs to steer patients, which changes negotiated prices.

8.3 Incremental contribution

The paper extends merger simulation to negotiated-price settings, provides a structural way to recover costs and bargaining weights from hospital claims data, and shows how patient insurance design interacts with antitrust policy. Its key substantive contribution is to demonstrate that hospital mergers can raise prices even when MCOs bargain actively, and that conduct remedies based on formal separate bargaining may fail.

8.4 Takeaway

In markets where prices are negotiated, concentration affects bargaining leverage, not just unilateral price-setting incentives. Antitrust analysis must therefore model the bargaining game and the downstream insurance environment. In hospital markets, patient coinsurance, MCO bargaining objectives, and hospital system ownership are all first-order determinants of prices.